

TMA4315 Exercise 1 Linear models for Gaussian data

Group 14

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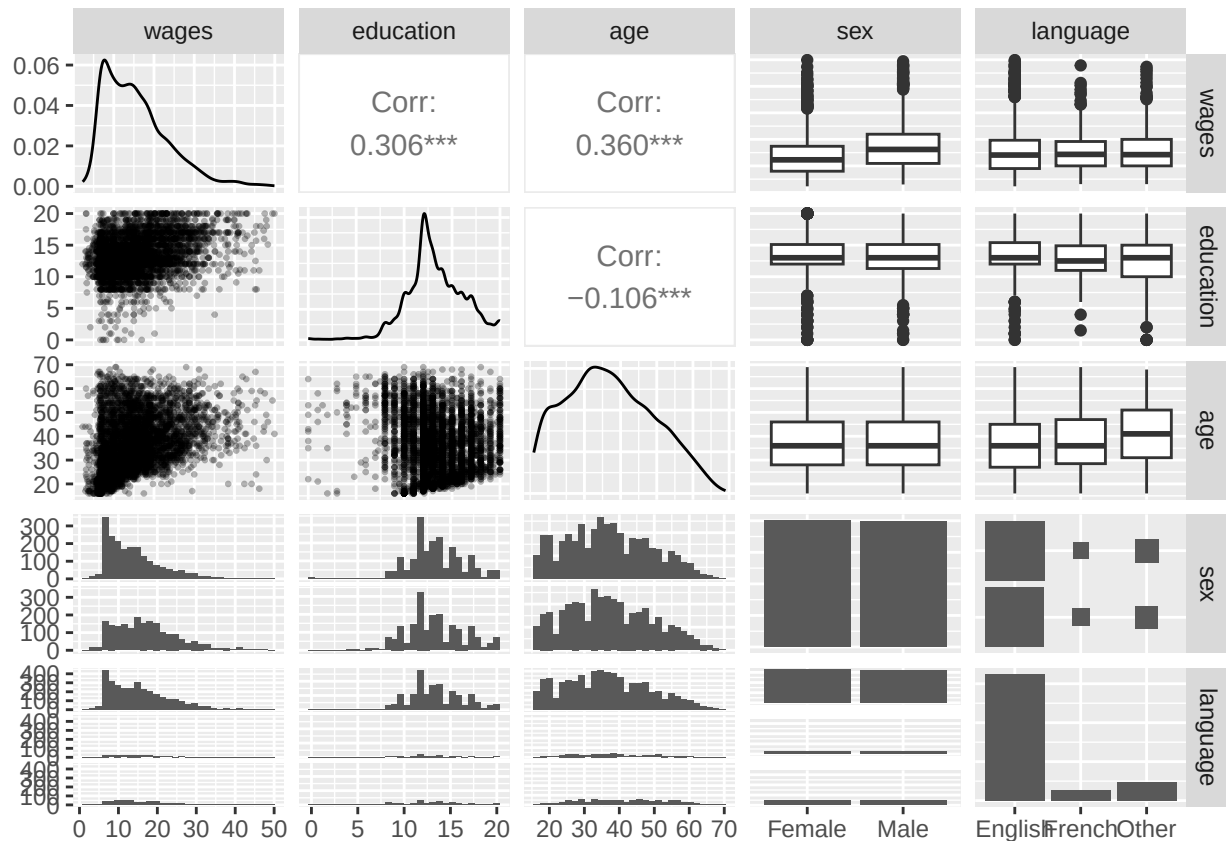
Part 1: Explanatory analysis of the dataset

First of all we import the dataset and prepare the data for the analysis.

```
library(car)
data(SLID, package = "carData")
SLID <- SLID[complete.cases(SLID), ]
```

Afterwards we draw a matrix of diagnostic plots of all the variables.

```
ggpairs(SLID, lower = list(continuous = wrap("points", alpha = 0.3, size=0.5)))
```



Looking at the plot, there appear to be some noteworthy relationships between various pairs of variables:

- **education** and **wages** have a moderate positive correlation of 0.306 that is, however, highly statistically significant. The same applies to **age** and **wages**.
- median wages are noticeably higher for male individuals compared to females, while they are effectively the same in relation to the language. This suggests that an individual's gender might be strongly related with their wage, while their language has a likely much weaker effect. In both cases, however, we can see that most wages are concentrated towards relatively low values, with the exception of a few, much higher outliers. This is further confirmed by observing the **wages** density plot, which is skewed to the right.
- **age** and **education** have a very weak, but highly statistically significant, negative correlation of -0.106. This means that these two variables are most likely uncorrelated.
- median education is effectively the same for male and female individuals, and very similar for individuals of different languages. This intuitively suggests that education is unaffected by these two variables. In both cases, the median education is relatively high, with the exception of a few outliers with a significantly lower value.
- a similar argument applies to the **age** variable, although it is worth noticing that the median age of individuals of **other** language appears to be higher.
- there does not seem to be anything noteworthy about the relationship between age and language. It can be noticed, however, that the split between male and female individuals seems almost equal, while the **language** variables is dominated by **English**.

To carry out a multiple linear regression analysis of y on the covariates $x_1, \dots, x_p$ we need the following

assumptions.

- **Linearity and additivity.** The expected response is a linear function of the regression parameters and the effects of the covariates add up:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \dots, n,$$

or in matrix form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$.

- **Homoscedasticity.** The errors have a constant variance, $\text{Var}(\varepsilon_i) = \sigma^2$ for all i , which does not depend on the covariates.
- **Uncorrelated errors.** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, so that $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$.
- **Full rank design.** The design matrix $\mathbf{X}$ has full column rank $p < n$, so that $\mathbf{X}^\top \mathbf{X}$ is invertible and $\hat{\boldsymbol{\beta}}$ is uniquely defined. The covariates are also assumed to be fixed (measured without error).
- **Normality.** For the *normal* linear regression model we additionally assume $\varepsilon_i \sim N(0, \sigma^2)$, independently. This is what makes the exact t - and F -tests valid. Importantly, it is not needed for $\hat{\boldsymbol{\beta}}$ to be unbiased or for the *asymptotic* z - and χ^2 -tests used in this assignment, which only rely on a large sample size.

Given the strong right-skewness of **wages** noted above, the normality assumption is the one most likely to be violated here, which is a further reason for preferring the asymptotic tests implemented in **mylm**.

Part 2: Linear regression with the mylm package

Statistical Formulas and Theory

The **mylm** package implements ordinary least squares (OLS) for a linear model $Y = X\boldsymbol{\beta} + \varepsilon$. The functions in the package are built upon the following formulas:

- **Coefficient estimates:** $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top Y$
- **Fitted values:** $\hat{Y} = X\hat{\boldsymbol{\beta}}$
- **Residuals:** $e = Y - \hat{Y}$
- **Residual Sum of Squares (SSE):** $SSE = e^\top e$
- **Degrees of Freedom (df):** $df = n - p$ (where p is the number of parameters).
- **Total Sum of Squares (SST):** $SST = \sum (y_i - \bar{y})^2$
- **Covariance matrix:** $\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (X^\top X)^{-1}$, with $\hat{\sigma}^2 = \frac{SSE}{n-p}$
- **Standard Errors (SE):** $\text{SE}(\hat{\beta}_j) = \sqrt{\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}})_{jj}}$
- **Asymptotic z-test:** $z_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$ with p-value $2\Phi(-|z_j|)$.
- **Asymptotic χ^2 -test:** $\chi^2 = \frac{SSE_{\text{reduced}} - SSE_{\text{full}}}{\hat{\sigma}_{\text{full}}^2}$
- **Coefficient of determination:** $R^2 = 1 - \frac{SSE}{SST}$

a) & b) Fitting the model and calculating standard errors

We fit a simple linear regression model to the data with **wages** as the response and **education** as the predictor.

```
model11 <- mylm(wages ~ education, data = SLID)
model11b <- lm(wages ~ education, data = SLID)

print.mylm(model11)
```

```
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
```

```
## Coefficients:
## (Intercept)    education
##    4.9716912    0.7923091
```

```
print(model1b)
```

```
##
## Call:
## lm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)    education
##    4.9717    0.7923
```

The output confirms that our mylm function yields the exact same coefficient estimates as the standard lm function.

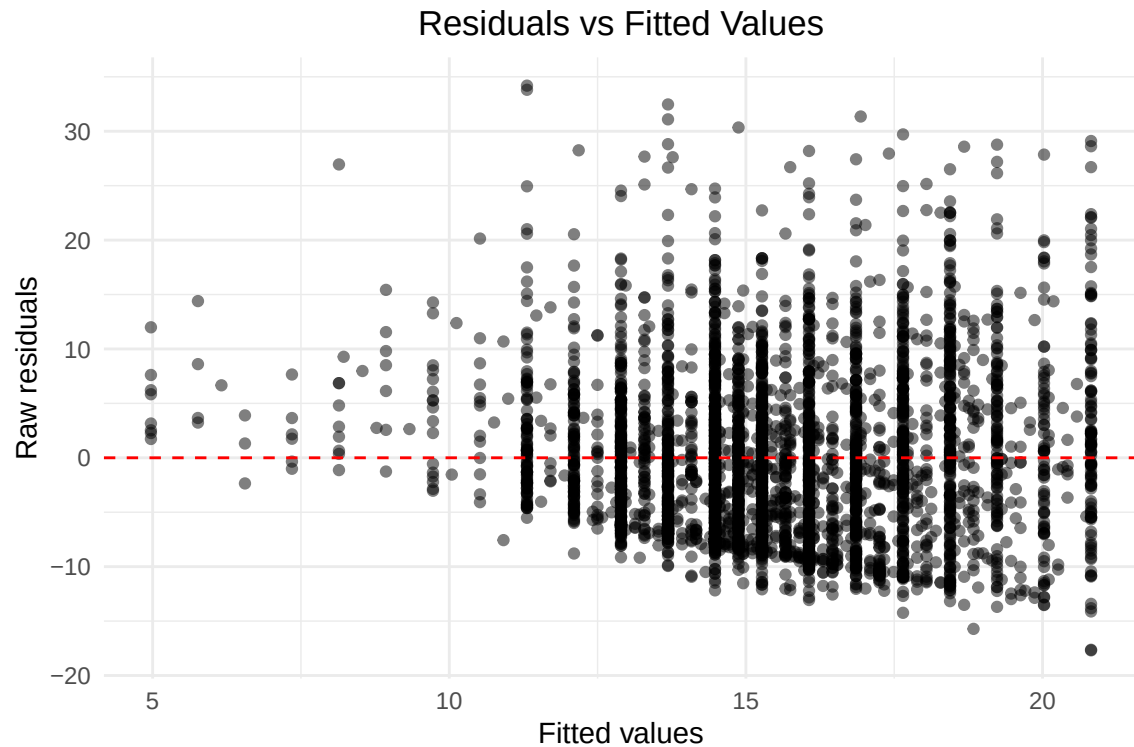
```
summary.mylm(model1)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.97169    0.53429  9.3052 < 2.2e-16 ***
## education    0.79231    0.03906 20.2842 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.093586
```

Interpretation of parameters: * **Intercept** ($\hat{\beta}_0 \approx 4.97$): The expected hourly wage for a person with 0 years of education. This is extrapolated outside the data range and holds no practical meaning since wages cannot be negative. * **Education** ($\hat{\beta}_1 \approx 0.79$): For each additional year of education, the expected hourly wage increases by 0.79 units. * **Significance (z-test)**: The z-value for education is 20.28, yielding a p-value strictly close to 0. We confidently reject the null hypothesis ($H_0 : \beta_1 = 0$); education is a highly significant predictor.

c) Residuals plot

```
plot.mylm(model1)
```



Comment on the plot: The residuals versus fitted values plot shows a funnel shape, meaning the variance of the residuals increases as the fitted values increase. This violates the assumption of homoscedasticity.

d) ANOVA and Chi-squared test

```
cat(sprintf(
  "SSE: %.2f\nDegrees of freedom: %d\nSST: %.2f",
  model1$sse, model1$df.residual, model1$sst
))

## SSE: 223694.27
## Degrees of freedom: 3985
## SST: 246790.47

sse_full <- model1$sse
df_full <- model1$df.residual
sst <- model1$sst

base_model <- mylm(wages ~ 1, data = SLID)
df_diff <- base_model$df.residual - model1$df.residual

chisq_val <- (sst - sse_full) / (sse_full / df_full)
p_val <- pchisq(chisq_val, df = df_diff, lower.tail = FALSE)

rm(base_model)

cat(sprintf(
  "SSE: %.2f\nDegrees of freedom: %d\nSST: %.2f\nTest degrees of freedom: %d\nChi-squared: %.2f\np-value: %.2f",
  sse_full, df_full, sst, df_diff, chisq_val, p_val
))
```

```
## SSE: 223694.27
## Degrees of freedom: 3985
## SST: 246790.47
## Test degrees of freedom: 1
## Chi-squared: 411.45
## p-value: 1.774739e-91
```

Relationship between tests: In simple linear regression, the χ^2 statistic with 1 degree of freedom is equal to the square of the z -statistic for the slope coefficient. From our summary, $z \approx 20.284$, and $z^2 \approx 411.45$, matching the χ^2 value in the ANOVA table. The critical value for a 5% significance level is $z_{0.025} \approx 1.96$ and $\chi^2_{1,0.05} \approx 3.84$, which our statistics comfortably exceed.

```
results_anova <- anova.mylm(model1)
```

```
## Analysis of Variance Table
## Response: wages
##           Df Sum sq X2 value Pr(>X2)
## education 1  23096.20 411.45  1.7747e-91
```

```
anova(model1b)
```

```
## Analysis of Variance Table
##
## Response: wages
##           Df Sum Sq Mean Sq F value    Pr(>F)
## education    1  23096 23096.2   411.45 < 2.2e-16 ***
## Residuals 3985  223694    56.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Finally, if we pass the model to our `anova.mylm()` function, it produces a sequential ANOVA table that matches the previous calculations. It displays the same 1 degree of freedom, Sum of Squares (23096.20), and asymptotic χ^2 test statistic (411.45), confirming that our implementation works correctly.

e) Coefficient of determination (R^2)

The coefficient of determination R^2 is **0.0936**.

Interpretation: Approximately **9.36%** of the total variability in hourly wages is explained by the variance in years of education. The remaining **90.64%** is unexplained by this model.

Part 3: Multiple linear regression

a) Fitting the multiple regression model

Because our `mylm` and `print.mylm` functions were strictly implemented using generalized matrix operations (specifically $\hat{\beta} = (X^T X)^{-1} X^T Y$), they automatically adapt to any number of predictors. No changes to the underlying R functions are necessary to handle multiple linear regression.

```
model2 <- mylm(wages ~ education + age, data = SLID)
print.mylm(model2)
```

```
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
## (Intercept)  education          age
```

```
## -6.0216529 0.9014644 0.2570898
```

b) Estimates, standard errors, and z-test

Similarly, our `summary.mylm` function uses the generalized covariance matrix formula $\hat{\sigma}^2(X^\top X)^{-1}$. Extracting the diagonal elements for the standard errors works flawlessly regardless of the model's dimensions.

```
summary.mylm(model2)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -6.0216529  0.6189238 -9.7292 < 2.2e-16 ***
## education    0.9014644  0.0357598 25.2089 < 2.2e-16 ***
## age          0.2570898  0.0089512 28.7212 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.249070
```

Interpretation of parameters:

- **Intercept** ($\hat{\beta}_0 \approx -6.02$): The expected hourly wage for a theoretical person with 0 years of education and an age of 0. Even though this doesn't make much sense practically, it is consistent with the regression model.
- **Education** ($\hat{\beta}_1 \approx 0.90$): This means that each additional year of education increases the hourly wage by 0.9, for equal age.
- **Age** ($\hat{\beta}_2 \approx 0.26$): This means that each additional year of age increases the hourly wage by 0.26, for equal education. Thus, we can see that education is valued more highly than age in determining wages.
- **Significance**: We can see from the summary that the p-values for age and education are far smaller than 5% (2.2e-16 for both), which means that age and education are highly significant predictors of wages.

c) Simple vs. Multiple Regression Estimates

The estimated parameter for `education` is not the same in the simple regression and in the multiple regression, because of the **omitted variable bias**. In the simple model, the coefficient of `education` has to take both its own effect on wages and the effect of the variable that we forgot (`age`). This happens because `age` and `education` are correlated in our dataset.

In the simple linear regression, the `education` coefficient absorbs both its direct effect on wages and the indirect effect of the omitted variable (`age`), since `age` and `education` are correlated (as seen in Part 1, $r = -0.106$). In the multiple linear regression, the coefficient isolates the **partial effect** of `education` on wages while explicitly controlling for `age`.

We can show these differences by running the individual models with `mylm` and comparing their coefficients:

```
# Fit the individual simple regressions
mod_edu <- mylm(wages ~ education, data = SLID)
mod_age <- mylm(wages ~ age, data = SLID)
```

```

# Display the isolated vs partial effects
cat("Simple regression education effect:  ", mod_edu$coefficients["education"], "\n")

## Simple regression education effect:      0.7923091
cat("Simple regression age effect:        ", mod_age$coefficients["age"], "\n\n")

## Simple regression age effect:            0.2331079
cat("Multiple regression education effect: ", model2$coefficients["education"], "\n")

## Multiple regression education effect:    0.9014644
cat("Multiple regression age effect:       ", model2$coefficients["age"], "\n")

## Multiple regression age effect:         0.2570898

```

Part 4: Testing the mylm package

Model 4

COMPARING PRINT OUTPUT

```

model4 <- mylm(wages ~ sex + age + language + I(education^2), data = SLID)
model4b <- lm(wages ~ sex + age + language + I(education^2), data = SLID)

print.mylm(model4)

##
## Call:
## mylm(formula = wages ~ sex + age + language + I(education^2),
##      data = SLID)
##
## Coefficients:
##      (Intercept)          sexMale          age languageFrench languageOther
##      -1.87553134      3.40870006      0.24862499      -0.07553202      -0.13454020
## I(education^2)
##      0.03481515

print(model4b)

```

```

##
## Call:
## lm(formula = wages ~ sex + age + language + I(education^2), data = SLID)
##
## Coefficients:
##      (Intercept)          sexMale          age languageFrench languageOther
##      -1.87553      3.40870      0.24862      -0.07553      -0.13454
## I(education^2)
##      0.03482

```

COMPARING SUMMARY OUTPUT

```
summary.mylm(model4)
```

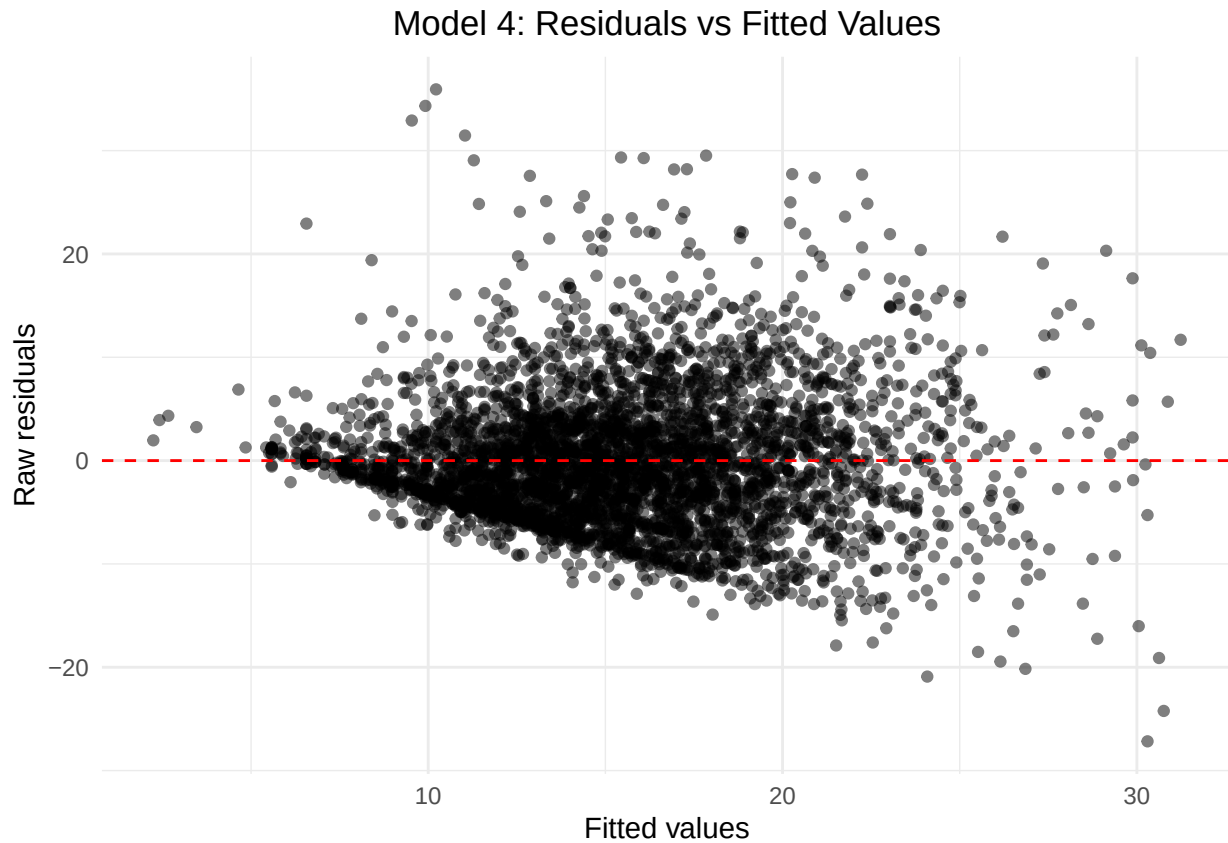
```
## Summary of object
##
## Call:
## mylm(formula = wages ~ sex + age + language + I(education^2),
##       data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.8755313  0.4403451 -4.2592 2.051e-05 ***
## sexMale       3.4087001  0.2084196 16.3550 < 2.2e-16 ***
## age           0.2486250  0.0086626 28.7009 < 2.2e-16 ***
## languageFrench -0.0755320  0.4251357 -0.1777  0.8590
## languageOther -0.1345402  0.3231525 -0.4163  0.6772
## I(education^2) 0.0348152  0.0012899 26.9907 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.302220
```

```
summary(model4b)
```

```
##
## Call:
## lm(formula = wages ~ sex + age + language + I(education^2), data = SLID)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.171  -4.276  -0.763   3.218  35.929
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.875531  0.440345  -4.259  2.1e-05 ***
## sexMale       3.408700  0.208420  16.355  < 2e-16 ***
## age           0.248625  0.008663  28.701  < 2e-16 ***
## languageFrench -0.075532  0.425136  -0.178  0.859
## languageOther -0.134540  0.323153  -0.416  0.677
## I(education^2) 0.034815  0.001290  26.991  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.577 on 3981 degrees of freedom
## Multiple R-squared:  0.3022, Adjusted R-squared:  0.3013
## F-statistic: 344.8 on 5 and 3981 DF, p-value: < 2.2e-16
```

MODEL PLOT

```
plot.mylm(model4, title = "Model 4: Residuals vs Fitted Values")
```



COMMENT

From the model's summary, we can see that **sexMale** is very positively correlated to **wages**, and this correlation is highly statistically significant, as its p-value is extremely small (less than $2.2e-16$). On the other hand, **age** and **education²** seem to have a very slight positive correlation with **wages**, as they have small positive coefficients. This is supported by the fact that their p-value is less than $2.2e-16$, meaning that these estimates are statistically significant. Finally, **languageFrench** and **languageOther** appear to be slightly negatively correlated to **wages**. However, given the high p-values of these estimates, we can't say with certainty if they are a result of random chance or noise. Since the coefficient of determination is around 0.3, we can conclude that this model explains about 30% of the variance of the dependent variable. The plot clearly shows that the assumption of homoscedasticity does not hold. Based on these results, we can conclude with relative confidence that a male individual is expected to earn about \$3.41 dollars per hour more than the female equivalent, with **female** being the baseline for the **sex** variable. At the same time, every year of age corresponds to an increase of around \$0.25 dollars per hour, while the **education²** coefficient indicates that there is a non-linear effect (similar to an acceleration) of education on wages. A possible change to the model could be the inclusion of the **education** parameter, in addition to the second order term, to have a more complete understanding of the effect of this variable on **wages**.

Model 5

COMPARING PRINT OUTPUT

```
model5 <- mylm(wages ~ language * age, data = SLID)
model5b <- lm(wages ~ language * age, data = SLID)
```

```
print.mylm(model5)
```

```
##
## Call:
## mylm(formula = wages ~ language * age, data = SLID)
##
## Coefficients:
##      (Intercept)      languageFrench      languageOther          age
##      6.55579376      2.86062508      0.84862130      0.24485160
## languageFrench:age languageOther:age
##      -0.08392752      -0.03701381
```

```
print(model5b)
```

```
##
## Call:
## lm(formula = wages ~ language * age, data = SLID)
##
## Coefficients:
##      (Intercept)      languageFrench      languageOther          age
##      6.55579      2.86063      0.84862      0.24485
## languageFrench:age languageOther:age
##      -0.08393      -0.03701
```

COMPARING SUMMARY OUTPUT

```
summary.mylm(model5)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ language * age, data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    6.555794   0.410681  15.9632 < 2e-16 ***
## languageFrench  2.860625   1.596072   1.7923  0.07309 .
## languageOther   0.848621   1.235182   0.6870  0.49206
## age            0.244852   0.010688  22.9101 < 2e-16 ***
## languageFrench:age -0.083928   0.040456  -2.0745  0.03803 *
## languageOther:age -0.037014   0.029340  -1.2615  0.20711
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.131170
```

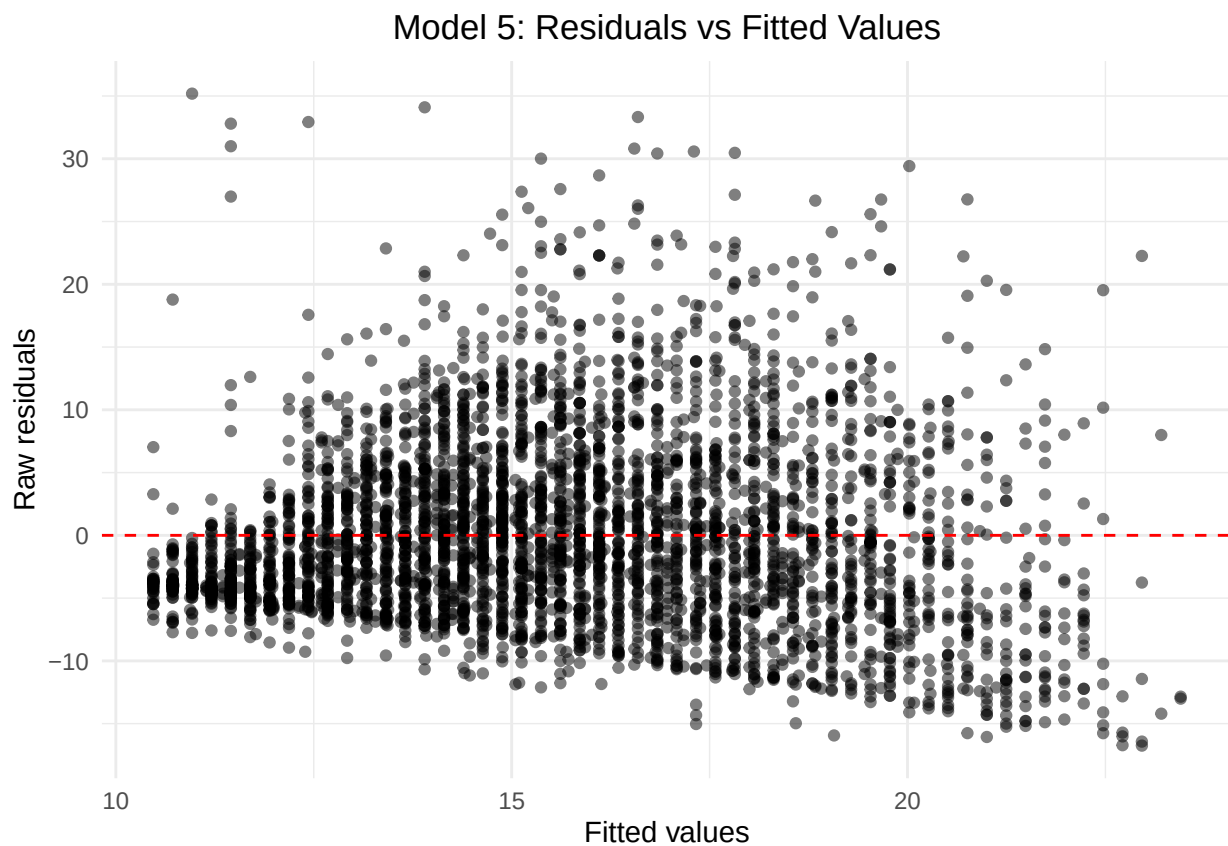
```
summary(model5b)
```

```
##
## Call:
## lm(formula = wages ~ language * age, data = SLID)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.751  -4.832  -1.412   3.938  35.187
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.55579    0.41068   15.963  <2e-16 ***
## languageFrench    2.86063    1.59607    1.792   0.0732 .
## languageOther     0.84862    1.23518    0.687   0.4921
## age               0.24485    0.01069   22.910  <2e-16 ***
## languageFrench:age -0.08393    0.04046   -2.075   0.0381 *
## languageOther:age -0.03701    0.02934   -1.262   0.2072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.339 on 3981 degrees of freedom
## Multiple R-squared:  0.1312, Adjusted R-squared:  0.1301
## F-statistic: 120.2 on 5 and 3981 DF,  p-value: < 2.2e-16
```

MODEL PLOT

```
plot.mylm(model5, title = "Model 5: Residuals vs Fitted Values")
```



COMMENT

Just like in the previous model, `age` is extremely statistically significant and lightly positively correlated

to `wages`. The interaction term `languageFrench:age` is also statistically significant, although less than `age`, and it shows a slightly negative coefficient. All the other terms, besides the intercept, are not statistically significant. The coefficient of determination here is about 0.13, meaning that this model only explains around 13% of the variance of the `wages` variable. Due to the presence of the interaction between `age` and `language` in the call, `age` now represents the wage increase per year of age for English speakers, `languageEnglish` being the baseline for the `language` variable. This means that `languageFrench` has a slightly weaker effect on `wages` than `languageEnglish`: a French speaker's wage increases less than an English speaker's for each year of age (specifically, \$0.08 less). Once again, the plot shows that homoscedasticity does not hold. One possible way to improve the model would be to compare it to a simpler model such as `mylm(formula = wages ~ language + age, data = SLID)`. Since most interaction terms seem to be weakly significant or not significant at all, it could be interesting to see if a simpler model is able to explain the data just as well as the current one. Alternatively, we could center the `age` variable in its mean, since wages at very low ages have little statistical meaning.

Model 6

COMPARING PRINT OUTPUT

```
model6 <- mylm(wages ~ education - 1, data = SLID)
model6b <- lm(wages ~ education - 1, data = SLID)

print.mylm(model6)

##
## Call:
## mylm(formula = wages ~ education - 1, data = SLID)
##
## Coefficients:
## education
## 1.146697

print(model6b)

##
## Call:
## lm(formula = wages ~ education - 1, data = SLID)
##
## Coefficients:
## education
## 1.147
```

COMPARING SUMMARY OUTPUT

```
summary.mylm(model6)

## Summary of object
##
## Call:
## mylm(formula = wages ~ education - 1, data = SLID)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## education 1.1466971  0.0087672 130.79 < 2.2e-16 ***
```

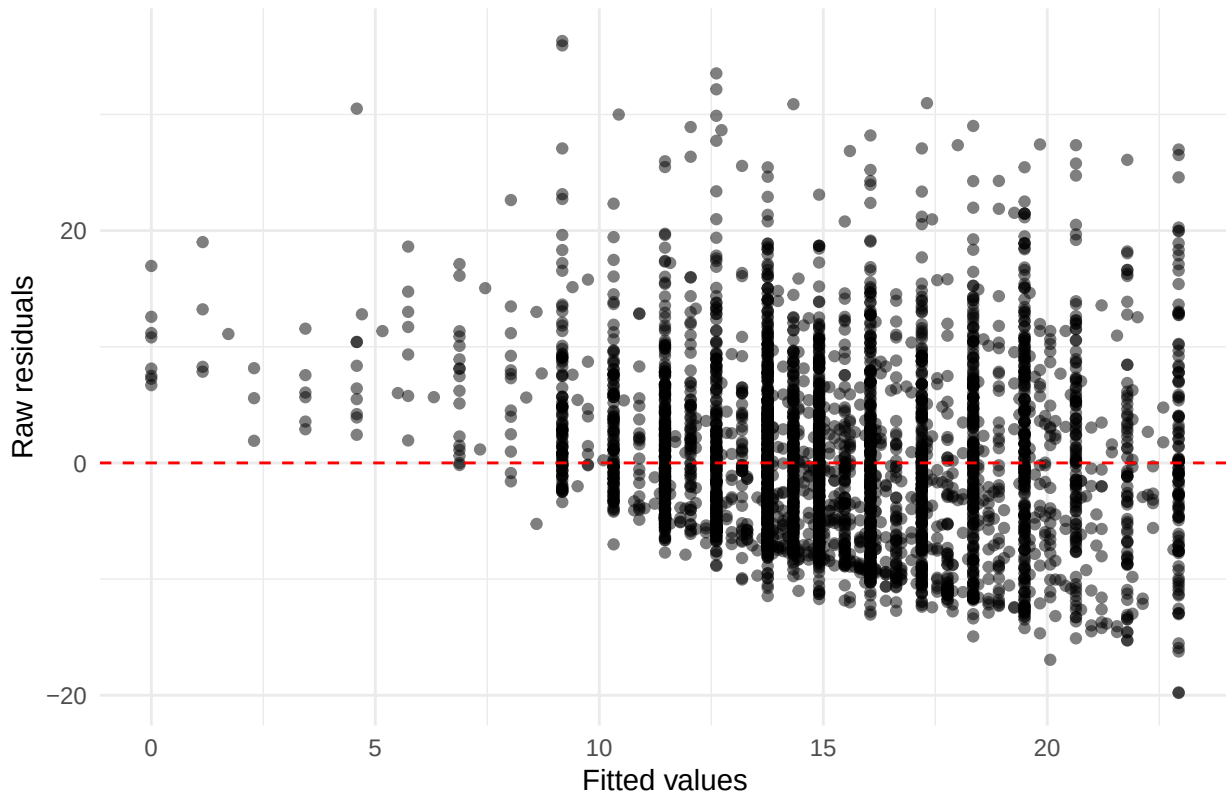
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.073892
summary(model6b)

##
## Call:
## lm(formula = wages ~ education - 1, data = SLID)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.804  -5.342  -0.662   4.465  36.326
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## education 1.146697    0.008767   130.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.572 on 3986 degrees of freedom
## Multiple R-squared:  0.811, Adjusted R-squared:  0.811
## F-statistic: 1.711e+04 on 1 and 3986 DF, p-value: < 2.2e-16
```

MODEL PLOT

```
plot.mylm(model6, title = "Model 6: Residuals vs Fitted Values")
```

Model 6: Residuals vs Fitted Values



COMMENT

The **education** coefficient is highly statistically significant, as its p-value is less than $2.2e-16$, and it has a positive effect on **wages**. Because the intercept was removed, this model's regression line passes through the origin. This can be interpreted as the model assuming that someone with 0 years of education earns exactly \$0.00 per hour. Since the **education** coefficient is almost 1.15, we can say that, according to this model, a person's hourly wage is expected to increase by about \$1.15 for each year of education they have. Once again, the plot shows that the homoscedasticity assumption does not hold. Clearly, however, removing the intercept from the model represents a very strong assumption: people with no education earn \$0.00 per hour. Clearly, this makes little logical sense in the real world, which means that this assumption is certainly wrong and the model can be improved by simply reinserting the intercept. Moreover, the R-squared is not defined when the intercept is removed, which is another reason to keep it in this case.

Final considerations and possible improvements

All three of these models suffer from two common issues: the lack of homoscedasticity and the right-skewness of **wages**, which is a symptom of the fact that the errors are likely not normally distributed. These problems can hardly be addressed just by modifying the models' linear structure. A better option would be to modify the dataset by applying some specific transformations. One option would be to replace **wages** with $\log(\text{wages})$, which could fix the right-skewness of the variable and, possibly, the lack of homoscedasticity of the residuals. However, an even more robust alternative is applying a Box-Cox transformation to the response variable. These transformations are commonly used to correct skewed data and often stabilize the variance of the residuals, allowing the homoscedasticity assumption to be satisfied.

Appendix

mylm.R

```
# Select Build, Build and reload to build and lode into the R-session.

mylm <- function(formula, data = list(), contrasts = NULL, ...){
  # Extract model matrix & responses
  mf <- model.frame(formula = formula, data = data)
  X <- model.matrix(attr(mf, "terms"), data = mf, contrasts.arg = contrasts)
  y <- model.response(mf)
  terms <- attr(mf, "terms")

  # Add code here to calculate coefficients, residuals, fitted values, etc...
  # and store the results in the list est
  df <- nrow(X) - ncol(X)
  xtx_inv <- solve(t(X)%*% X)
  beta_hat <- drop(xtx_inv %*% t(X) %*% y)
  fitted_values <- X %*% beta_hat
  res <- y - fitted_values

  sse <- sum(res^2)
  sst <- sum((y - mean(y))^2)
  sigma2 <- sse / df
  vcov_mat <- as.numeric(sigma2) * xtx_inv
  r_squared <- 1 - (sse / sst)

  est <- list(terms = terms, model = mf)
  est$coefficients <- beta_hat
  est$fitted.values <- fitted_values
  est$residuals <- res
  est$df.residual <- df
  est$sse <- as.numeric(sse)
  est$sst <- as.numeric(sst)
  est$vcov <- vcov_mat
  est$r_squared <- as.numeric(r_squared)

  # Store call and formula used
  est$call <- match.call()
  est$formula <- formula

  # Set class name. This is very important!
  class(est) <- 'mylm'

  # Return the object with all results
  return(est)
}

print.mylm <- function(object, ...){
  # Code here is used when print(object) is used on objects of class "mylm"
  # Useful functions include cat, print.default and format
  cat('\nCall:\n')
  print(object$call)
}
```



```

    cat('\nCoefficients:\n')
    print(object$coefficients)
}

summary.mylm <- function(object, ...){
  # Code here is used when summary(object) is used on objects of class "mylm"
  # Useful functions include cat, print.default and format
  cat('Summary of object\n')
  se <- sqrt(diag(object$vcov))
  z_val <- object$coefficients / se
  p_val <- 2 * pnorm(-abs(z_val))
  coef_table <- cbind(Estimate = object$coefficients, `Std. Error` = se, `z value` = z_val, `Pr(>|z|)` = p_val)
  cat('\nCall:\n')
  print(object$call)
  cat('\nCoefficients:\n')
  printCoefmat(coef_table, P.values = TRUE, has.Pvalue = TRUE)
  cat(sprintf('\nMultiple R-squared:  %f\n', object$r.squared))
}

plot.mylm <- function(object, title = "Residuals vs Fitted Values",...){
  # Code here is used when plot(object) is used on objects of class "mylm"
  library(ggplot2)
  df_plot <- data.frame(fitted = object$fitted.values, residuals = object$residuals)

  p <- ggplot(df_plot, aes(x = fitted, y = residuals)) +
    geom_point(alpha = 0.5) +
    geom_hline(yintercept = 0, col = "red", linetype = "dashed") + theme_minimal() +
    labs(title = title, x = "Fitted values", y = "Raw residuals") +
    theme(plot.title = element_text(hjust = 0.5))
  print(p)
}

# This part is optional! You do not have to implement anova
anova.mylm <- function(object, ...){
  # Code here is used when anova(object) is used on objects of class "mylm"

  # Components to test
  comp <- attr(object$terms, "term.labels")

  # Name of response
  response <- deparse(object$terms[[2]])

  # Fit the sequence of models
  txtFormula <- paste(response, "~", sep = "")
  model <- list()
  for(numComp in 1:length(comp)){
    if(numComp == 1){
      txtFormula <- paste(txtFormula, comp[numComp])
    }
    else{
      txtFormula <- paste(txtFormula, comp[numComp], sep = "+")
    }
  }
}

```

```

}
formula <- formula(txtFormula)
model[[numComp]] <- mylm(formula = formula, data = object$model)
}

# Print Analysis of Variance Table
cat('Analysis of Variance Table\n')
cat(c('Response: ', response, '\n'), sep = '')
cat('      Df  Sum sq X2 value Pr(>X2)\n')
base_formula <- formula(paste(response, "~ 1"))
base_model <- mylm(formula = base_formula, data = object$model)
prev_sse <- base_model$sse
prev_df <- base_model$df.residual
for(numComp in 1:length(comp)){
  # Add code to print the line for each model tested
  curr_sse <- model[[numComp]]$sse
  curr_df <- model[[numComp]]$df.residual

  df_diff <- prev_df - curr_df
  ss_diff <- prev_sse - curr_sse

  chisq_val <- ss_diff / (object$sse / object$df.residual)
  p_val <- pchisq(chisq_val, df = df_diff, lower.tail = FALSE)

  cat(sprintf("%-9s %-3d %-7.2f %-8.2f %-8.4e\n",
              comp[numComp], df_diff, ss_diff, chisq_val, p_val))

  prev_sse <- curr_sse
  prev_df <- curr_df
}

return(model)
}

```